EXP NO :-10
PROGRAM:-

```
clc;
clear;
close all;

% Values from Experiment 1 of the PDF
Fs = 1000;
Fp = 100;
Fst = 150;
Ap = 1;
Ast = 40;

% Butterworth LPF coefficients from Experiment 1
[N,Wn] = buttord(Fp/(Fs/2),Fst/(Fs/2),Ap,Ast);
[b,a] = butter(N,Wn,'low');

% PDF values
Bcoef = 8;
Bprod = 16;

% Coefficient quantization
Sc = 2^(Bcoef-1);
bq = round(b*Sc)/Sc;
aq = round(a*Sc)/Sc;

disp('Original coefficients:');
disp(b);
```

```
disp(a);

disp('8-bit Quantized coefficients:');
disp(bq);
disp(aq);

% Unit impulse
x = [1 zeros(1,99)];

% Ideal output
y_ideal = filter(b,a,x);

% Product quantization
Sp = 2^(Bprod-1);
yq = zeros(size(x));

for n = 1:length(x)

    temp = 0;

    for k = 1:length(bq)
        if n-k+1 > 0
            temp = temp + bq(k)*x(n-k+1);
        end
    end

    for k = 2:length(aq)
        if n-k+1 > 0
            temp = temp - aq(k)*yq(n-k+1);
```

```
        end
    end

    yq(n) = round(temp*Sp)/Sp;
end

figure;
stem(y_ideal,'filled');
hold on;
stem(yq,'r');
grid on;
legend('Ideal','Quantized');
title('Coefficient and Product Quantization');
xlabel('Sample');
ylabel('Amplitude');

clc;
clear;
close all;

% Experiment 1 values from PDF
Fs = 1000;
Fp = 100;
Fst = 150;
Ap = 1;
Ast = 40;

[N,Wn] = buttord(Fp/(Fs/2),Fst/(Fs/2),Ap,Ast);
[b,a] = butter(N,Wn,'low');
```

```
% PDF coefficient word length = 8 bits
B = 8;
S = 2^(B-1);

bq = round(b*S)/S;
aq = round(a*S)/S;

% Frequency response
[H,w] = freqz(b,a,1024,Fs);
[Hq,~] = freqz(bq,aq,1024,Fs);

figure;
plot(w,20*log10(abs(H)+eps));
hold on;
plot(w,20*log10(abs(Hq)+eps),'r--');
grid on;
legend('Ideal','8-bit Quantized');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Finite Word-Length Effect on Filter Response');

% Unit impulse output
x = [1 zeros(1,99)];

y = filter(b,a,x);
yq = filter(bq,aq,x);

figure;
```

```
stem(y,'filled');
hold on;
stem(yq,'r');
grid on;
legend('Ideal','8-bit');
xlabel('Sample');
ylabel('Amplitude');
title('Effect of Finite Word Length on Output');

clc;
clear;
close all;

% Experiment 1 values
Fs = 1000;
Fp = 100;
Fst = 150;
Ap = 1;
Ast = 40;

[N,Wn] = buttord(Fp/(Fs/2),Fst/(Fs/2),Ap,Ast);
[b,a] = butter(N,Wn,'low');

% PDF values
Bcoef = 8;
Bprod = 16;

Sc = 2^(Bcoef-1);
Sp = 2^(Bprod-1);
```

```
bq = round(b*Sc)/Sc;
aq = round(a*Sc)/Sc;

% First establish internal state using unit impulse
N1 = 50;
x = [1 zeros(1,N1-1)];

y = zeros(1,N1);

for n = 1:N1
   temp = 0;

   for k = 1:length(bq)
      if n-k+1 > 0
         temp = temp + bq(k)*x(n-k+1);
      end
   end

   for k = 2:length(aq)
      if n-k+1 > 0
         temp = temp - aq(k)*y(n-k+1);
      end
   end

   y(n) = round(temp*Sp)/Sp;
end

% Zero-input condition
```

```
N2 = 100;
x0 = zeros(1,N2);
y0 = zeros(1,N2);

% Continue from last filter states
for n = 1:N2
    temp = 0;

    for k = 2:length(aq)
        if n-k+1 > 0
            temp = temp - aq(k)*y0(n-k+1);
        end
    end

    y0(n) = round(temp*Sp)/Sp;
end

figure;
stem(y0,'filled');
grid on;
xlabel('Sample');
ylabel('Amplitude');
title('Zero-Input Limit-Cycle Test');

% Overflow-related test
x_over = ones(1,N2);
y_over = zeros(1,N2);

for n = 1:N2
```

```
    temp = 2*x_over(n);

    if n > 1
        temp = temp - aq(2)*y_over(n-1);
    end

    temp = round(temp*Sp)/Sp;

    % Normalized full-scale range
    y_over(n) = min(max(temp,-1),1);
end

figure;
plot(y_over);
grid on;
xlabel('Sample');
ylabel('Amplitude');
title('Overflow Limit-Cycle / Saturation Test');

clc;
clear;
close all;

% Experiment 1 values
Fs = 1000;
Fp = 100;
Fst = 150;
Ap = 1;
Ast = 40;
```

```
[N,Wn] = buttord(Fp/(Fs/2),Fst/(Fs/2),Ap,Ast);
[b,a] = butter(N,Wn,'low');

% PDF values
Bcoef = 8;
Bprod = 16;

Sc = 2^(Bcoef-1);
Sp = 2^(Bprod-1);

bq = round(b*Sc)/Sc;
aq = round(a*Sc)/Sc;

% Unit impulse
Nsample = 1000;
x = [1 zeros(1,Nsample-1)];

y = filter(b,a,x);

% Quantized filter
yq = zeros(size(x));

for n = 1:length(x)

    temp = 0;

    for k = 1:length(bq)
        if n-k+1 > 0
```

```
            temp = temp + bq(k)*x(n-k+1);
        end
    end

    for k = 2:length(aq)
        if n-k+1 > 0
            temp = temp - aq(k)*yq(n-k+1);
        end
    end

    yq(n) = round(temp*Sp)/Sp;
end

% Quantization error
eq = yq-y;

MSE = mean(eq.^2);

Psignal = mean(y.^2);
Perror = mean(eq.^2);

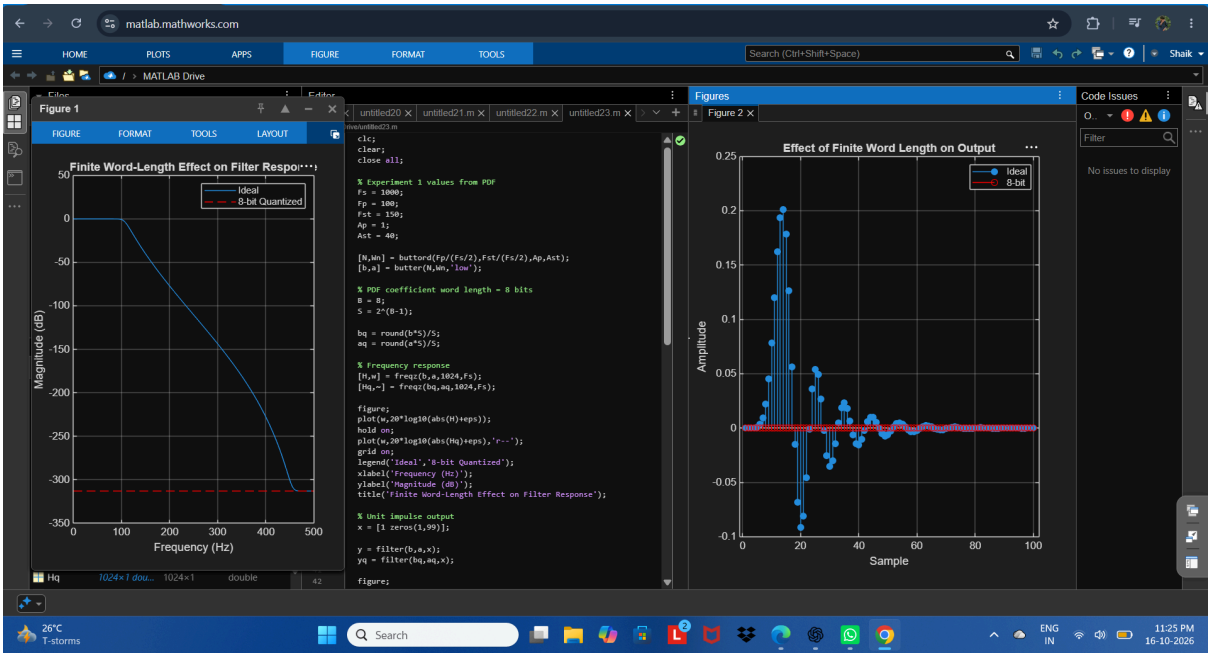
SQNR = 10*log10(Psignal/Perror);

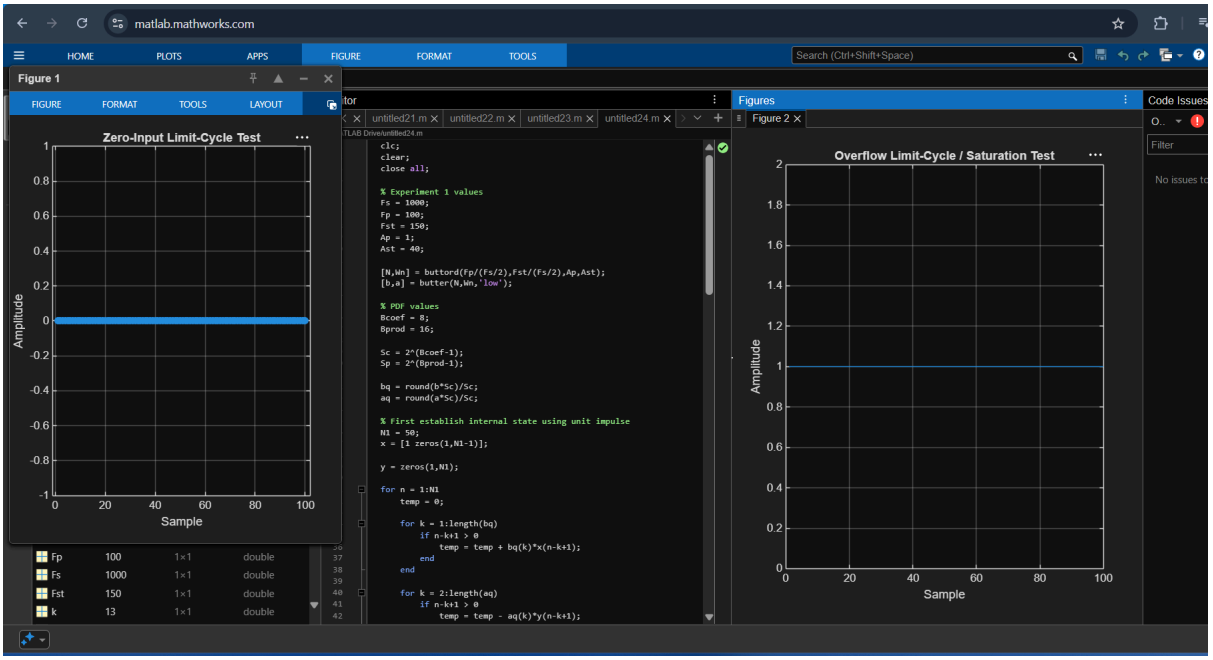
fprintf('MSE = %e\n',MSE);
fprintf('SQNR = %.4f dB\n',SQNR);

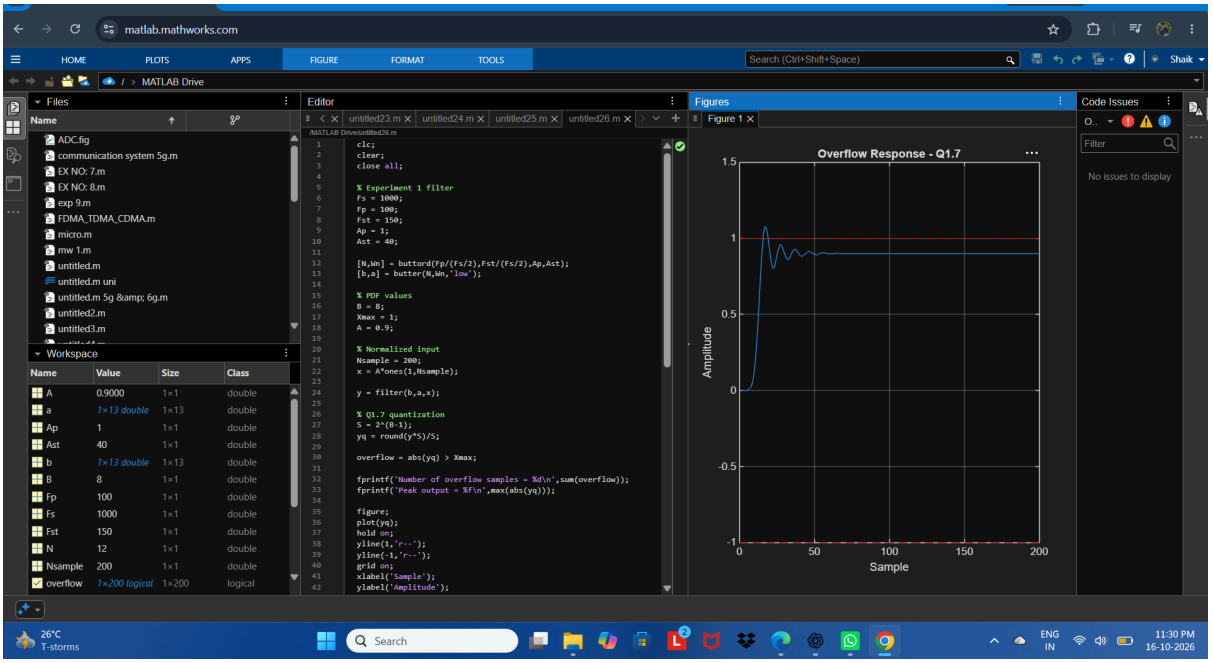
figure;
plot(eq);
grid on;
```

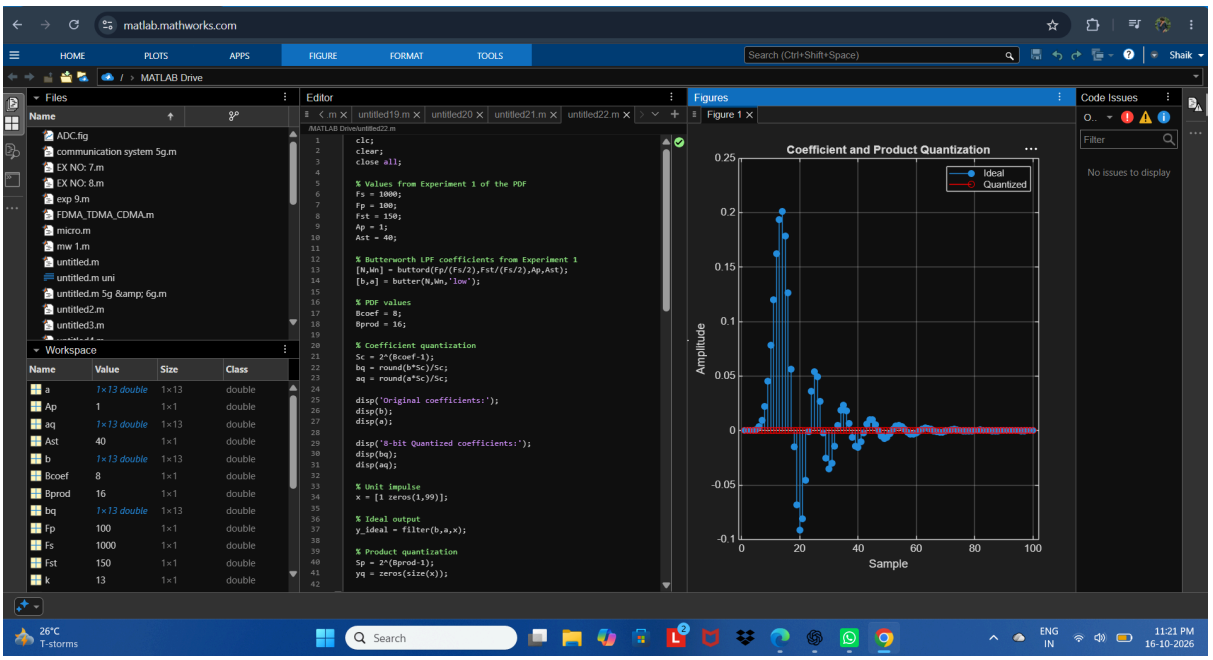
```
xlabel('Sample');
ylabel('Error');
title('Quantization Noise / Error');

figure;
plot(y);
hold on;
plot(yq,'r--');
grid on;
legend('Ideal','Quantized');
xlabel('Sample');
ylabel('Amplitude');
title('Performance Comparison');
```


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Figure 1
Finite Word-Length Effect on Filter Respo
Ideal
8-bit Quantized
Magnitude (dB)
Frequency (Hz)
Figures
Figure 2
Effect of Finite Word Length on Output
Amplitude
Sample
Code Issues
No issues to display
% Experiment 1 values from PDF
Fs = 1000;
Fp = 100;
Fst = 150;
Ap = 1;
Ast = 40;
% PDF coefficient word length = 8 bits
B = 8;
% Frequency response
% Unit impulse output


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Figure 1
Zero-Input Limit-Cycle Test
Amplitude
Sample
Figures
Figure 2
Overflow Limit-Cycle / Saturation Test
Amplitude
Sample
Code Issues
% Experiment 1 values
Fs = 1000;
Fp = 100;
Fst = 150;
Ap = 1;
Ast = 40;
% PDF values
Bcoef = 8;
Bprod = 16;
% First establish internal state using unit impulse
N1 = 50;
Fp 100 1×1 double
Fs 1000 1×1 double
Fst 150 1×1 double
k 13 1×1 double


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HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
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Shaik
MATLAB Drive
Files
Name
ADC.fig
communication system 5g.m
EX NO: 7.m
EX NO: 8.m
exp 9.m
FDMA_TDMA_CDMA.m
micro.m
mw 1.m
untitled.m
untitled.m uni
untitled.m 5g & 6g.m
untitled2.m
untitled3.m
Workspace
Name
Value
Size
Class
A 0.9000 1×1 double
a 1×13 double 1×13 double
Ap 1 1×1 double
Ast 40 1×1 double
b 1×13 double 1×13 double
B 8 1×1 double
Fp 100 1×1 double
Fs 1000 1×1 double
Fst 150 1×1 double
N 12 1×1 double
Nsample 200 1×1 double
overflow 1×200 logical 1×200 logical
Editor
untitled23.m
untitled24.m
untitled25.m
untitled26.m
clc;
clear;
close all;
% Experiment 1 filter
Fs = 1000;
Fp = 100;
Fst = 150;
Ap = 1;
Ast = 40;
[N,Wn] = buttord(Fp/(Fs/2),Fst/(Fs/2),Ap,Ast);
[b,a] = butter(N,Wn,'low');
% PDF values
B = 8;
Xmax = 1;
A = 0.9;
% Normalized input
Nsample = 200;
x = A*ones(1,Nsample);
y = filter(b,a,x);
% Q1.7 quantization
S = 2^(B-1);
yq = round(y*S)/S;
overflow = abs(yq) > Xmax;
fprintf('Number of overflow samples = %d\n',sum(overflow));
fprintf('Peak output = %f\n',max(abs(yq)));
figure;
plot(yq);
hold on;
yline(1,'r--');
yline(-1,'r--');
grid on;
xlabel('Sample');
ylabel('Amplitude');
Figures
Figure 1
Overflow Response - Q1.7
Amplitude
Sample
Code Issues
No issues to display
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HOME
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Shaik
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ADC.fig
communication system 5g.m
EX NO: 7.m
EX NO: 8.m
exp 9.m
FDMA_TDMA_CDMA.m
micro.m
mw 1.m
untitled.m
untitled.m uni
untitled.m 5g & 6g.m
untitled2.m
untitled3.m
Workspace
Name
Value
Size
Class
a 1×13 double 1×13 double
Ap 1 1×1 double
aq 1×13 double 1×13 double
Ast 40 1×1 double
b 1×13 double 1×13 double
Bcoef 8 1×1 double
Bprod 16 1×1 double
bq 1×13 double 1×13 double
Fp 100 1×1 double
Fs 1000 1×1 double
Fst 150 1×1 double
k 13 1×1 double
Editor
untitled19.m
untitled20
untitled21.m
untitled22.m
clc;
clear;
close all;
% Values from Experiment 1 of the PDF
Fs = 1000;
Fp = 100;
Fst = 150;
Ap = 1;
Ast = 40;
% Butterworth LPF coefficients from Experiment 1
[N,Wn] = buttord(Fp/(Fs/2),Fst/(Fs/2),Ap,Ast);
[b,a] = butter(N,Wn,'low');
% PDF values
Bcoef = 8;
Bprod = 16;
% Coefficient quantization
Sc = 2^(Bcoef-1);
bq = round(b*Sc)/Sc;
aq = round(a*Sc)/Sc;
disp('Original coefficients:');
disp(b);
disp(a);
disp('8-bit Quantized coefficients:');
disp(bq);
disp(aq);
% Unit impulse
x = [1 zeros(1,99)];
% Ideal output
y_ideal = filter(b,a,x);
% Product quantization
Sp = 2^(Bprod-1);
yq = zeros(size(x));
Figures
Figure 1
Coefficient and Product Quantization
Ideal
Quantized
Amplitude
Sample
Code Issues
No issues to display
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